

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

May 2028
College Park, MD

Experience

Amazon Web Services

Software Development Engineer Intern

Sep 2026 – Dec 2026
Redmond, WA

- Writing Java to tune upper/lower bounds on EC2 capacity reserved for Amazon Redshift operations like patching, backtesting policies to balance availability against cost

Capital One

Software Engineering Intern

Jun 2026 – Aug 2026
Richmond, VA

- Rebuilt a legacy test-data workflow builder in React/TypeScript + Electron, adding a React-Flow editor and Jest-tested validation; **90%** of 30+ surveyed developers preferred it to the old tool
- Precomputed a tree-style search index over popular workflows from a **270K+ row** PostgreSQL table (Python/SQLAlchemy, Redis cache) to minimize DB queries under access limits
- Cut LLM workflow generation time **65%** to under 15s by decomposing a Claude skill into sub-skills backed by Python search/validation tools, guided by CloudWatch + OpenSearch latency logs

UMD App Development Contracting (Client: Amazon Leo)

Software Engineer (Contract)

Jan 2026 – Present
College Park, MD

- Shipped a space-research tool used by Amazon Leo's public-policy team, now maintained by Leo engineers; deployed on AWS Fargate (Docker) via GitHub Actions and CDK (TypeScript)
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Automated Bedrock source-relevancy model fine-tuning on an EventBridge schedule, with decision gates that skip redundant retraining; added X-Ray tracing and SNS failure alerts

UMD Institute for Advanced Computer Studies (UMIACS)

Machine Learning Research Intern

Dec 2025 – Present
College Park, MD

- Built a Python pipeline for PhysTwin (physics-based digital twin; PhD-led team targeting ICLR) running AnyGrasp on multi-camera images and rendering grasps as Open3D overlays for a VLM, plus prompt tuning
- Shortened research-environment setup time **30%** with uv dependency pinning and Docker; added psutil time/memory profiling and Pytest suites to find bottlenecks

TerpLabs

Co-Founder, Software Engineer

Jun 2025 – Present
College Park, MD

- Co-founded a **110+ member** campus product studio that has shipped 5 live products
- Wrote Prisma SQL queries and Redis query caching to reduce DB load and latency for Tortuga, a class scheduler that drew **20,000+ visits** in its first 30 days
- Building a RAG meal-suggestion feature for the official UMD app (GPT-4o mini, ChromaDB + SentenceBERT search); **93%** of generated meals rated satisfactory by internal testers

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Engineered a C++17 order book and matching engine that replayed **50.9M** real IEX DEEP+ messages with no resync, matching 99,122 of 99,123 fills exactly and 100% of top-of-book prices
- Benchmarked B+ tree vs LSM skip-list log stores on identical 2M-record workloads: **12.8×** read throughput and 22× faster range summaries for 27% more disk
- Sustained **1.26M events/sec** end-to-end with 686.7 ns/event matching at a 1.40 GHz base clock, 1.41× the cost of a passive mirror book for deciding every fill itself

Technical Skills

Languages: Python, Java, C++, C, TypeScript/JavaScript, SQL

Cloud (AWS): Lambda, Kinesis, Fargate/ECS, API Gateway, EventBridge, Bedrock, CDK, CloudWatch, X-Ray, SNS

Web & Data: React, Electron, Node.js, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

AI/ML & Tools: PyTorch, RAG, Open3D, Docker, GitHub Actions, Linux, Pytest, Jest, uv, perf

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)